

Lambda

- Aws lambda is compute services that lets you run code without provisioning or managing server
- With Aws lambda , you can run code for virtually any type of application or backend services all with zero administration
- Aws lambda manage all the administration it manages –
- Provisioning and capacity of the compute fleet that offers a balance of memory cpu network and other Resources
- Server and OS maintenance
- High availability and automatic scaling
- Monitoring fleet health
- Deploying your code
- Monitoring and logging your lambda function
- Aws lambda runs your code on a high availability compute infrastructure



- Aws lambda run your code on a high availability compute infrastructure
- Aws lambda execute your code only when needed and scale automatically from a few request per day to thousands second
- You pay only for the compute time you consume no charge your code in the form of one or more lambda function to Aws lambda , in one of the language that aws support (currently node.js , java , PowerShell, c# , ruby python etc and services can run the code on your behalf
- Typically the lifecycle for an Aws lambda based application include authoring code deploying code to Aws lambda and then monitoring and troubleshooting
- This is in exchange for flexibility which means you can't log into compute instance or customize the operating system or language runtime
- If you do want to manage your own compute you can use ec2 or elastic beanstalk

How Lambda works ?

- First you upload your code to lambda in one or more lambda function
- Aws lambda will then execute the code in your behalf
- After the code is invoked lambda automatically take care of provisioning and managing the required servers
- Aws lambda is platform as a services
- It support only limited language like node-js java c# ruby go and PowerShell
- Write your code and push the code into Aws lambda
- You can't log into compute instance choose customized o.s or language platform

Important Term used in Lambda

Functions :

- A function is a resource that you can invoke to run your code in Aws lambda
- A function has code that process Events and a runtime that passes request and response between lambda and the function code

Runtime :

- Lambda runtime allow function is different language to run in the same base execution environment
- The runtime site in between lambda runtime site in between the lambda services and your function code, relaying invocation , events context information and response between the two

Event :

- Events is a Json formatted document that contains data for a function to process

Event source/Trigger :

- An Aws services as Amazon sns or custom services that triggers your function and executes its logic

Downstream:

- An Aws services such as DynamoDB tables or s3 buckets that your lambda function calls once it is triggers

Concurrency :

- No. of Request that your function is serving in any given time